

SymPy Bug Card

Bug 3: solveset gives nan as a secant zero

Task. Solve the secant equation over the complex domain:

$$\sec(x) = 0.$$

SymPy's answer.

$$\text{solveset}(\sec(x), x, \mathbb{C}) = \{\text{nan}\}.$$

Correct answer.

$$\emptyset.$$

Explanation. For finite complex z , $\sec(z) = 1/\cos(z)$ is never zero. The reported candidate is not a complex solution; $\sec(\text{nan}) = \text{nan}$, not 0.

Likely root cause. The diagnosis traces the result to `_invert_trig_hyp_complex` in `sympy/solvers/solveset.py`, which applies `asec(0)` while inverting $\sec(x) = 0$ and lacks a reciprocal-trig range guard excluding zero. This is a new instance of a known failure mode.

Artifact (GitHub). [bug-report/bug-003-solveset-gives-nan-as-a-secant-zero/](#)